

torch.kthvalue#

<https://pytorch.org/docs/stable/generated/torch.kthvalue.html>

Description:

Returns a namedtuple (values, indices) where values is the kth smallest element of each row of the input tensor in the given dimension dim. And indices is the index location of each element found. If dim is not given, the last dimension of the input is chosen. If keepdim is True, both the values and indices tensors are the same size as input, except in the dimension dim where they are of size 1. Otherwise, dim is squeezed (see `torch.squeeze()`), resulting in both the values and indices tensors having 1 fewer dimension than the input tensor. When input is a CUDA tensor and there are multiple valid kth values, this function may nondeterministically return indices for any of them.

Function Signature

```
torch.kthvalue(input, k, dim=None, keepdim=False, *,  
out=None)#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (tuple, optional) – the output tuple of (Tensor, LongTensor) can be optionally given to be used as output buffers

Code Examples

```
>>> x = torch.arange(1., 6.)
>>> x
tensor([ 1.,  2.,  3.,  4.,  5.])
>>> torch.kthvalue(x, 4)
torch.return_types.kthvalue(values=tensor(4.),
indices=tensor(3))

>>> x=torch.arange(1.,7.).resize_(2,3)
>>> x
tensor([[ 1.,  2.,  3.],
        [ 4.,  5.,  6.]])
>>> torch.kthvalue(x, 2, 0, True)
torch.return_types.kthvalue(values=tensor([[4., 5., 6.]]),
indices=tensor([[1, 1, 1]]))
```

Notes & Warnings

NOTE When input is a CUDA tensor and there are multiple valid k th values, this function may nondeterministically return indices for any of them.

Detailed Documentation

Documentation

Returns a namedtuple (values, indices) where values is the k th smallest element of each row of the input tensor in the given dimension dim. And indices is the index location of each element found.

If dim is not given, the last dimension of the input is chosen.

If keepdim is True, both the values and indices tensors are the same size as input, except in the dimension dim where they are of size 1. Otherwise, dim is squeezed (see `torch.squeeze()`), resulting in both the values and indices tensors having 1 fewer dimension than the input tensor.

When input is a CUDA tensor and there are multiple valid k th values, this function may nondeterministically return indices for any of them.

input (Tensor) – the input tensor.

k (int) – k for the k -th smallest element

dim (int, optional) – the dimension to find the k th value along

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out (tuple, optional) – the output tuple of (Tensor, LongTensor) can be optionally given to be used as output buffers

